BUSINESS INTELLIGENCE AND TOOLS

EXPERIMENT 10

Report Scheduling, Automation and Alerts using Microsoft Power BI

Stepwise Procedure

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Aim

To understand and design Business Intelligence deployment, monitoring, backup, and recovery strategies using Microsoft Power BI for ensuring availability, reliability, and continuity of BI solutions.

Introduction

Business Intelligence (BI) solutions are useful only when their reports, dashboards, and analytical outputs are available to the people who need them. After development, a BI solution must be deployed, monitored, maintained, and protected. Deployment makes BI content available to end users. Monitoring checks the operational health of the solution, including refresh status, usage, performance, and user activity. Backup and restore practices protect BI assets against accidental deletion, corruption, and system failures.

This experiment considers Global Superstore as the business scenario. The organization has developed reports, dashboards, analytical solutions, security policies, and automated reporting mechanisms. The management team now requires an organizational deployment approach, a monitoring strategy, and documented backup and recovery procedures.

Tools / Software Required

- Microsoft Power BI Desktop
- Microsoft Power BI Service
- Global Superstore Dataset.xlsx
- Microsoft Excel
- Microsoft PowerPoint (optional)

Business Scenario and Scope

The student acts as a BI Solution Architect and designs a complete deployment and maintenance strategy for the Global Superstore BI environment. The scope includes deployment planning, workspace and access planning, monitoring strategy design, backup and restore planning, and analysis of a dashboard deletion incident. This report documents a proposed strategy; it does not claim that a live organizational deployment or an actual recovery operation was performed.

Part A – BI Deployment Plan

The first activity is to identify and inventory the BI assets created in earlier experiments. The inventory should include reports, dashboards, datasets, and workspaces that are intended for deployment.

BI Asset	Purpose / Contents	Intended Users	Deployment Consideration
Reports	Detailed analysis and visual summaries created in Power BI	Business users and department teams	Publish the approved report and verify its data source.
Dashboards	High-level view of important business indicators	Management and executives	Confirm key tiles, links, and access permissions.
Datasets	Data model and data used by reports	BI team and report consumers indirectly	Check refresh configuration and data accuracy.
Workspaces	Organized area for BI content and collaboration	BI administrators and authorized users	Set role-based access and governance rules.

1 Deployment Objectives

- Make approved reports and dashboards available to their intended users.
- Organize BI assets so that they can be managed and maintained efficiently.
- Provide access according to user responsibilities.
- Support reliable and timely access to updated information.
- Establish operational ownership for deployment and maintenance.

2 Proposed Deployment Sequence

1. Review and approve the reports, dashboards, and datasets produced in previous experiments.
2. Confirm that the Global Superstore data source and report content are correct.
3. Organize the content in an appropriate Power BI workspace.
4. Assign users and workspace roles according to their responsibilities.
5. Publish or share the approved BI content through the organization's Power BI environment.
6. Verify that intended users can access the content and that reports display correctly.

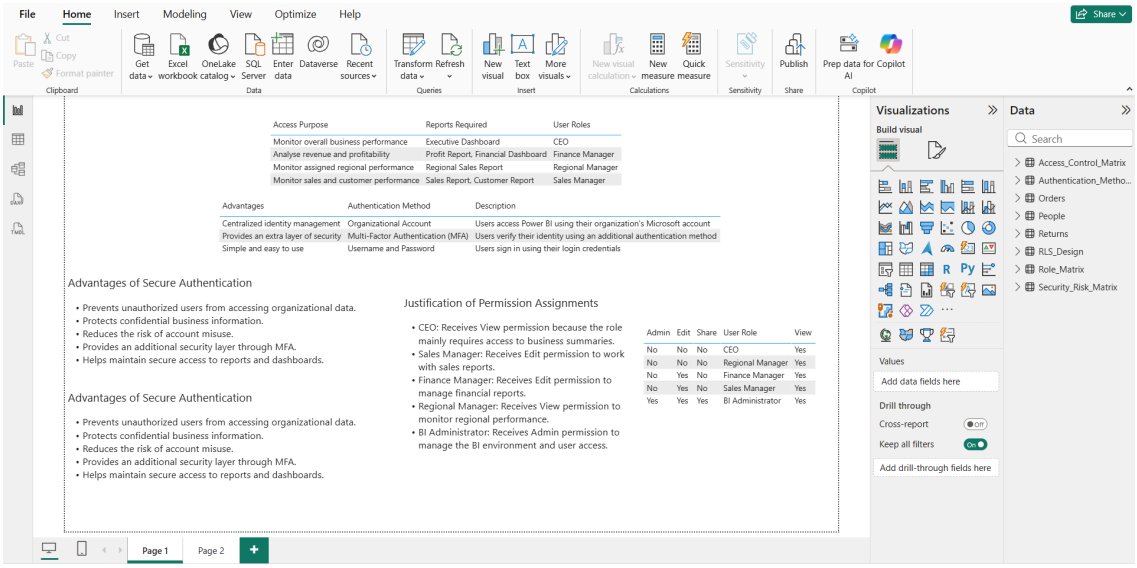


Figure: Page 1 of the Power BI report showing authentication, user roles, and permission planning.

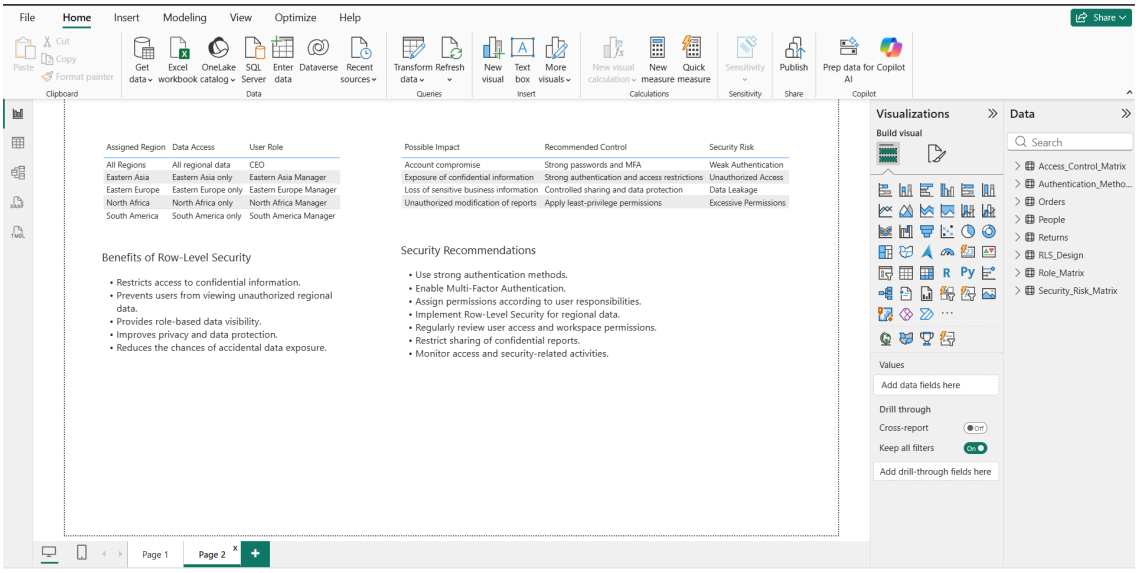


Figure: Page 2 of the Power BI report showing regional data access and security recommendations.

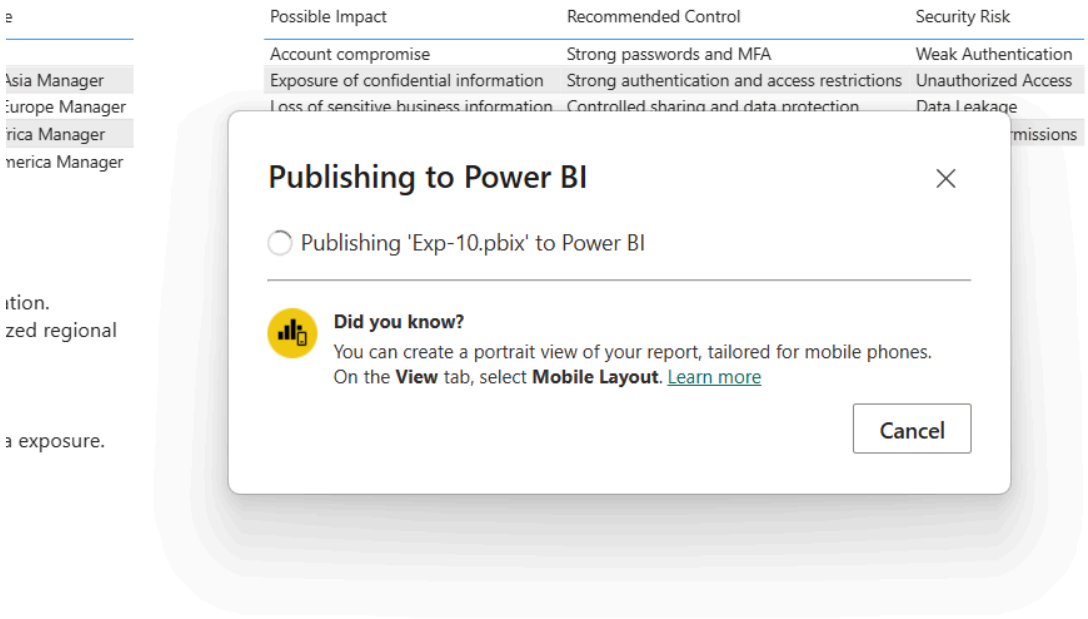


Figure: Power BI Desktop publishing the Exp-10 report to Power BI Service.

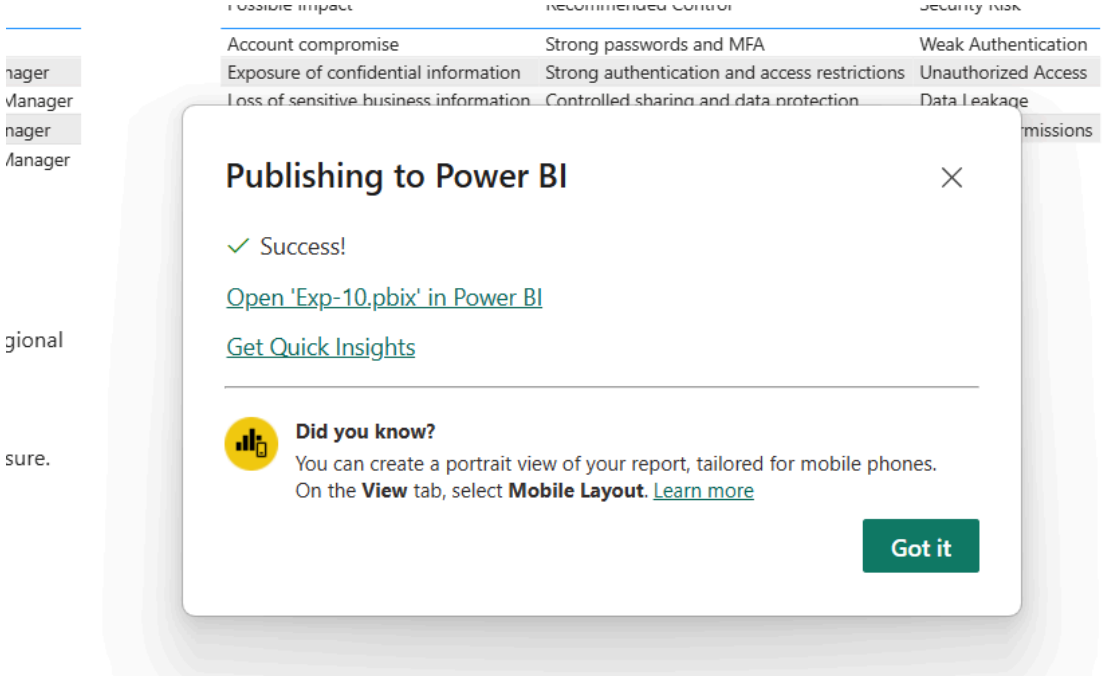


Figure: Successful publication of Exp-10 to Power BI Service.

Part B – Workspace and Access Planning

A workspace structure provides an organized location for related BI assets. Access should be assigned using defined roles so that users receive only the permissions required for their work.

Role	Main Responsibility	Access Planning
Administrator	Manages workspace settings, access, and overall governance.	Assign only to designated BI administrators.

Member	Collaborates on workspace content and manages shared BI assets.	Assign to users responsible for ongoing collaboration.
Contributor	Creates and updates reports and other workspace content.	Assign to BI developers and content creators.
Viewer	Views and interacts with published BI content.	Assign to business users and report consumers.

1 Workspace Governance Guidelines

- Use clear and consistent workspace and report names.
- Assign a responsible owner to each important dataset, report, and dashboard.
- Grant access according to job responsibilities and review access periodically.
- Limit administrative permissions to authorized personnel.
- Publish approved content and avoid unnecessary duplicate copies.
- Document important workspace changes and ownership details.

2 Workspace Structure (Conceptual)

Workspace Area	Content	Typical Access
Development	Work-in-progress datasets and reports	Administrator, Member, Contributor
Testing / Validation	Reports and datasets being checked before release	Administrator, Member, designated reviewers
Production	Approved reports and dashboards for business use	Administrator, authorized content managers, Viewer users

The Development–Testing–Production arrangement is a proposed organizational structure for this report. The experiment asks for a workspace structure design but does not prescribe a specific number or naming convention for workspaces.

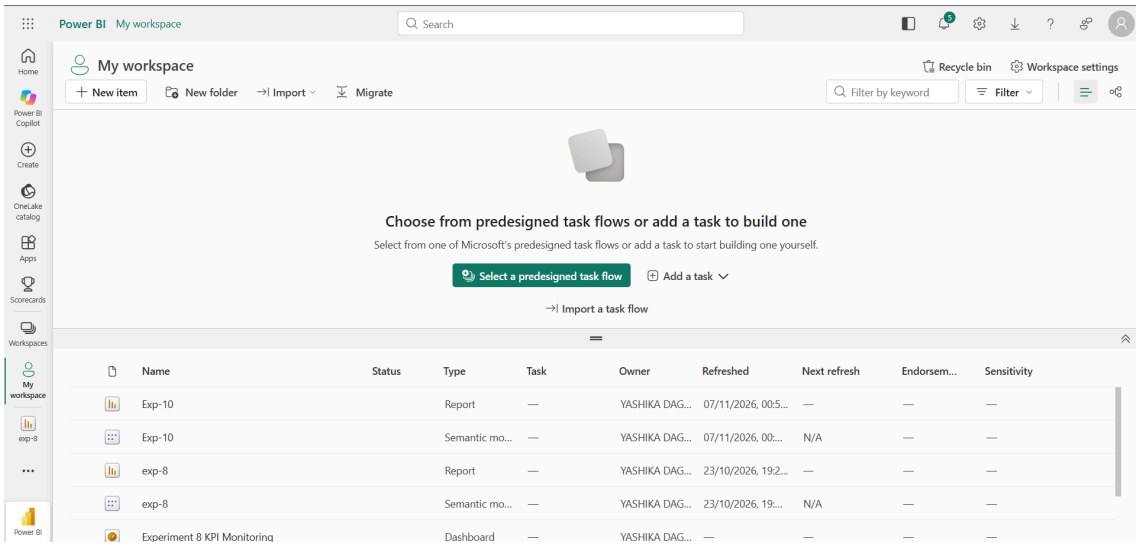


Figure: Exp-10 report and semantic model listed in the Power BI Service workspace.

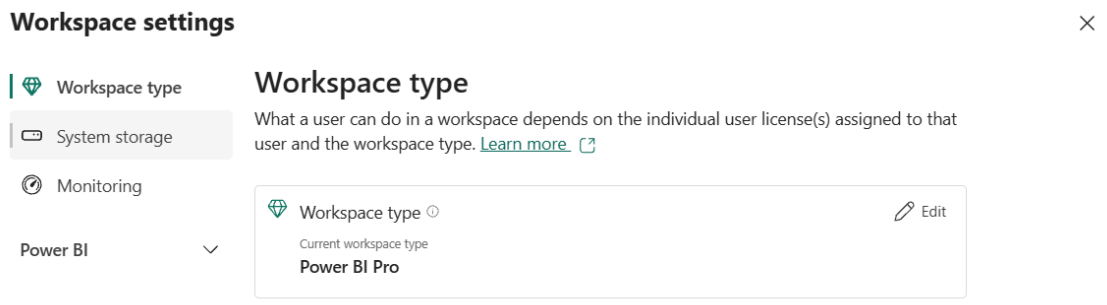


Figure: Workspace settings showing the current workspace type as Power BI Pro.

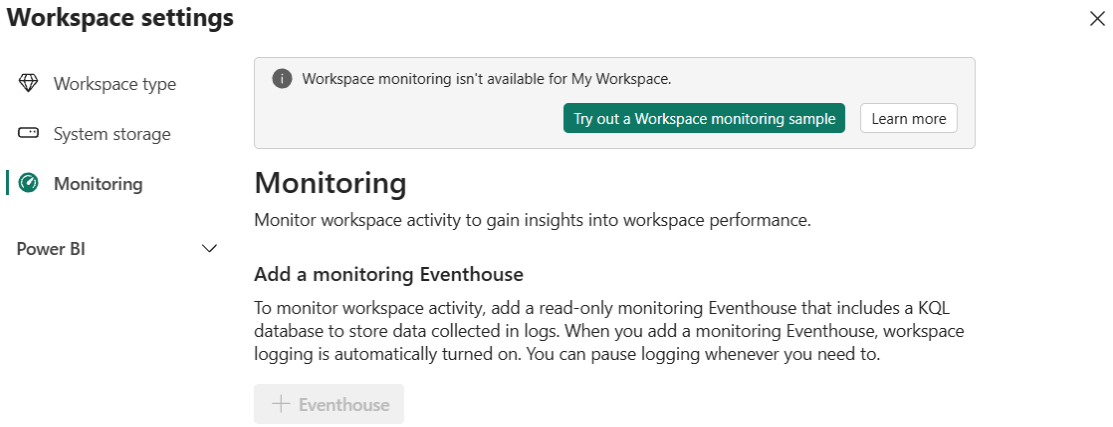


Figure: Monitoring settings page; workspace monitoring is not available for My Workspace.

Part C – Monitoring Strategy

Monitoring helps determine whether the deployed BI solution is operating correctly and whether users can access useful information. The experiment identifies four key monitoring areas: dataset refresh status, dashboard usage statistics, report performance, and user activity.

Metric / Area	What to Check	Suggested Frequency	Responsible Personnel
Dataset refresh status	Success or failure of scheduled refresh and freshness of data	Daily and after a reported failure	Dataset owner / BI administrator
Dashboard usage statistics	Views and usage trends for important dashboards	Weekly	BI administrator / report owner
Report performance	Loading responsiveness and slow visuals or pages	Weekly and after major changes	Report developer
User activity monitoring	Access patterns and relevant user activity	Weekly or as per governance policy	BI administrator

The frequencies and role assignments above are proposed planning values for the experiment. They should be adjusted to the organization's operational requirements.

1 Monitoring and Escalation Process

7. Review the monitoring matrix at the defined frequency.
8. Record failed refreshes, unusual usage changes, performance concerns, and access issues.
9. Assign each issue to the responsible dataset owner, report developer, or BI administrator.
10. Investigate the cause and apply the required correction.
11. Recheck the affected report or dataset and document the resolution.

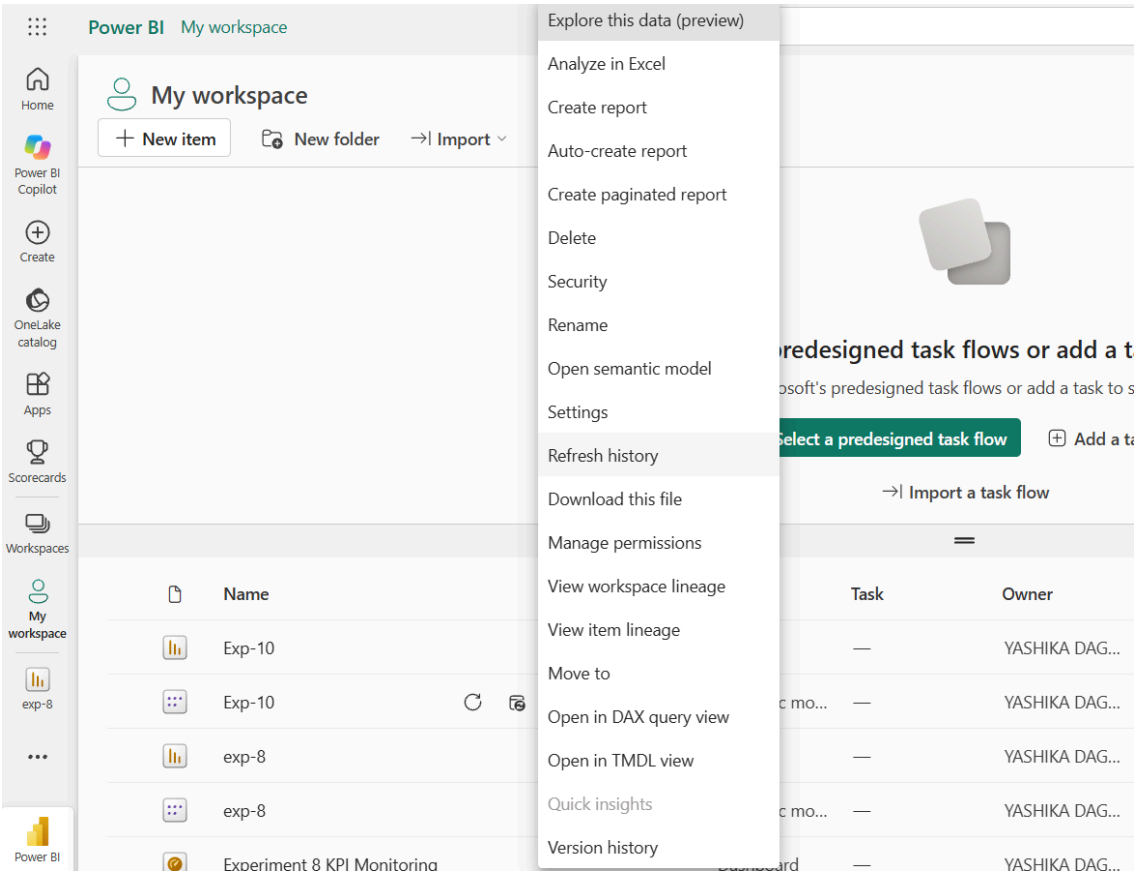


Figure: Refresh history option available from the Power BI Service item menu.

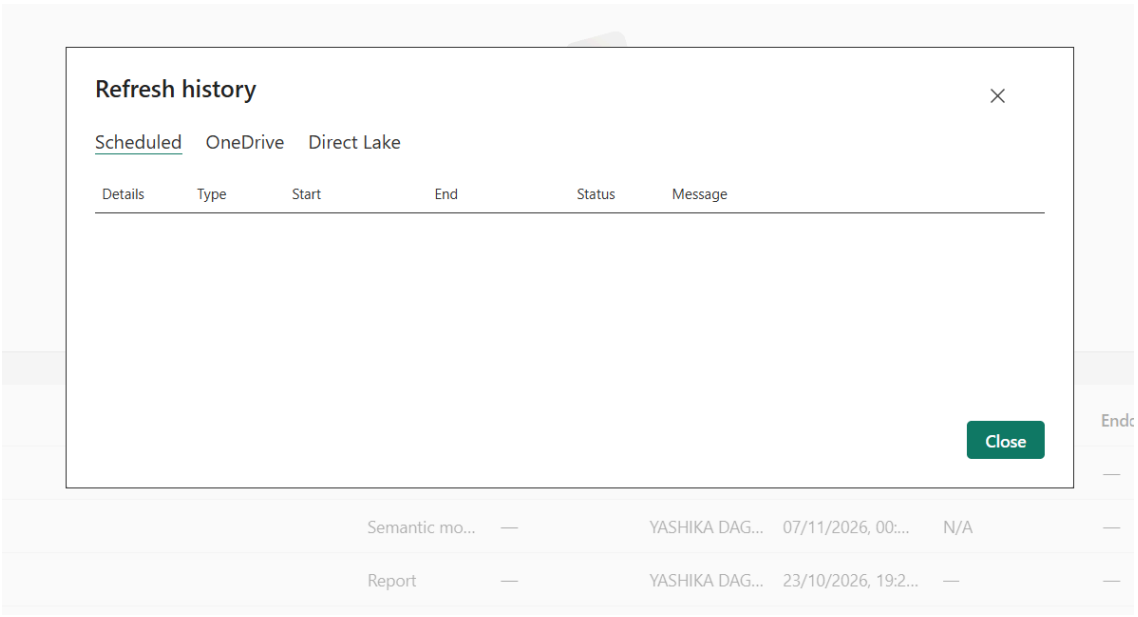


Figure: Refresh history view before a refresh record is displayed.

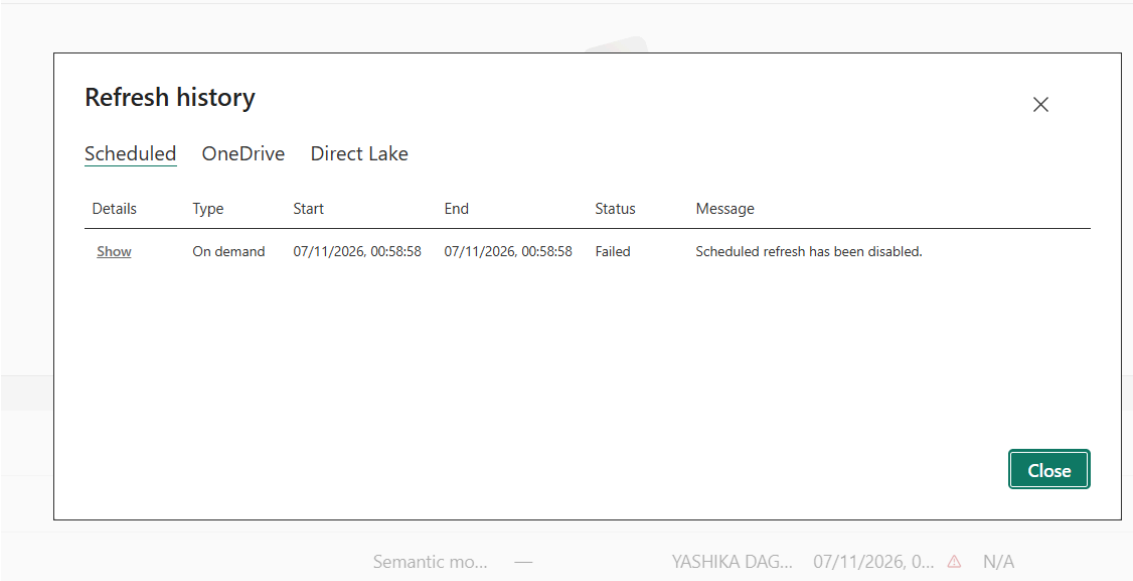


Figure: Refresh history indicating that scheduled refresh has been disabled.

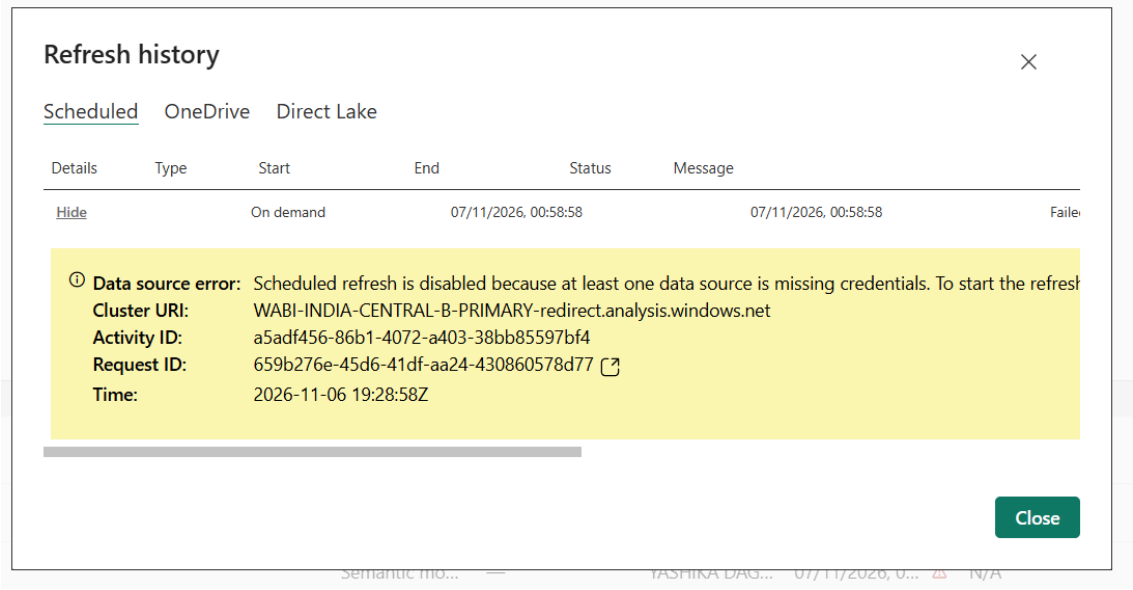


Figure: Refresh failure details identifying missing data-source credentials.

Part D – Backup and Restore Plan

Backup and restore mechanisms protect BI assets from accidental deletion, corruption, or system failure. The experiment gives examples of a daily dataset backup, weekly report backup, and monthly workspace archive.

Component	Proposed Backup Frequency	Backup Record / Verification
Dataset and related source data	Daily	Record date, owner, storage location, and whether the backup can be opened or restored.

Report files	Weekly	Keep a dated copy of approved report files and verify file availability.
Workspace archive / documentation	Monthly	Archive relevant workspace inventory, ownership, access details, and supporting documentation.

1 Restore Procedure

12. Identify the affected BI component and determine the impact on users.
13. Notify the BI administrator and the owner of the affected asset.
14. Locate the latest valid and verified backup or retained project file.
15. Restore or republish the affected report/dashboard and reconnect the required dataset where necessary.
16. Validate the data, visuals, filters, refresh behavior, and access permissions.
17. Inform the affected users after service is restored.
18. Document the incident, restoration source, actions taken, and any preventive measures.

2 Backup Governance

- Maintain a backup register with asset name, backup date, owner, and storage location.
- Keep backup copies in an approved and access-controlled location.
- Check backup availability periodically instead of assuming that a completed copy is usable.
- Retain more than one dated version where organizational policy permits.
- Restrict deletion and modification rights for backup repositories.

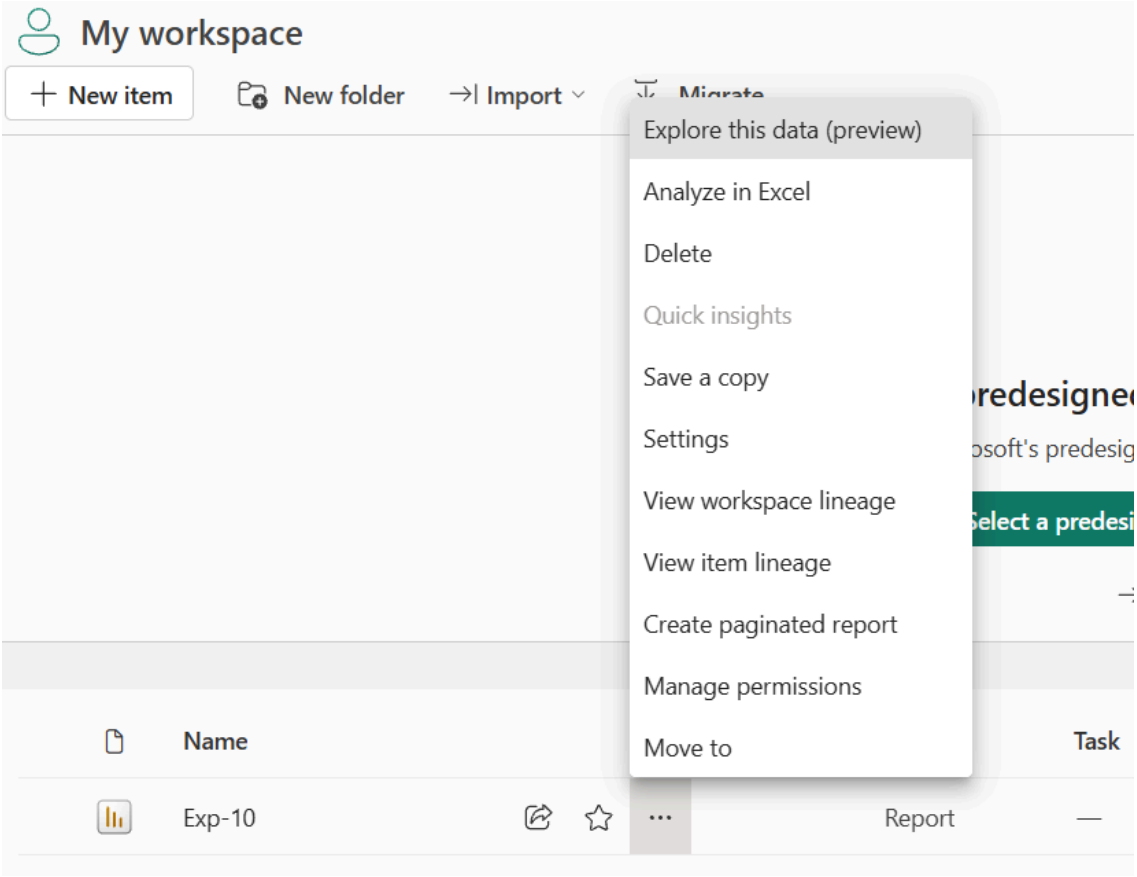


Figure: Power BI Service item menu showing the Save a copy option.

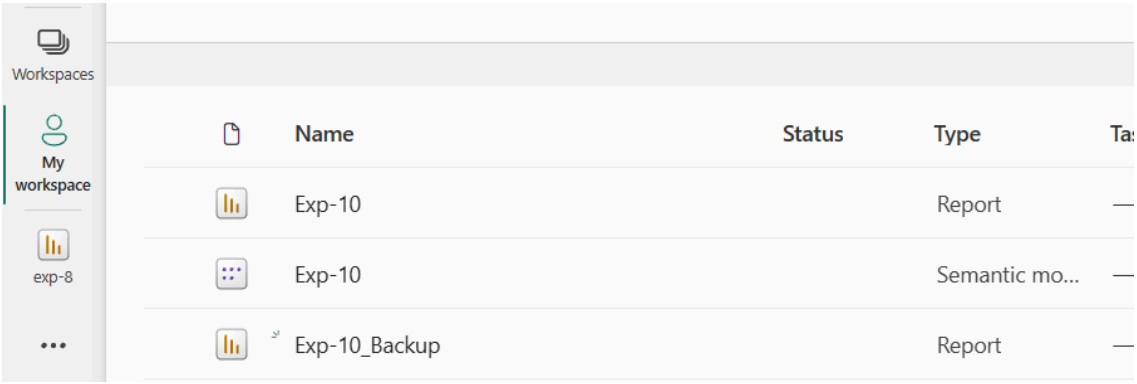


Figure: Backup copy of the Exp-10 report appearing in the workspace.

Part E – Disaster Recovery Scenario

Scenario

A critical executive dashboard is accidentally deleted one day before a management meeting.

1 Impact

The deletion may prevent executives from viewing important business information during the meeting. It can interrupt decision-making and reduce confidence in the availability of the BI solution. The impact depends on whether the dashboard, its underlying report, and its dataset are still available.

2 Recovery Actions

19. Confirm which dashboard was deleted and identify its workspace, owner, and dependent report/dataset.
20. Inform the BI administrator and relevant stakeholders about the incident.
21. Locate the most recent valid dashboard/report backup or retained Power BI project file.
22. Restore or republish the content to the appropriate workspace and confirm its connection to the correct dataset.
23. Check the dashboard's visuals, filters, data values, links, and permissions.
24. Verify that the underlying dataset is available and refreshed as expected.
25. Notify management when the dashboard is available and provide an alternative approved report or export if restoration is still in progress.

3 Business Continuity Measures

- Keep scheduled and versioned backups of critical reports and supporting files.
- Maintain an inventory of critical dashboards, owners, dependencies, and intended users.
- Define who is authorized to initiate recovery and communicate with stakeholders.
- Validate backup copies periodically through a restore check.
- Keep an approved alternative view or report available for time-critical management information.

Result

A complete BI deployment, monitoring, backup, and recovery strategy was designed for the Global Superstore scenario. The plan identifies BI assets and intended users, proposes workspace roles and governance guidelines, defines a monitoring matrix, establishes a backup schedule, and documents restoration and business continuity actions for the accidental deletion of a critical executive dashboard.

Learning Outcomes

- Analyze deployment requirements for Business Intelligence solutions.
- Design workspace structures and governance models for BI systems.
- Evaluate monitoring requirements for deployed BI solutions.
- Develop backup and recovery strategies for business continuity.
- Design complete deployment and maintenance plans for organizational BI environments.

Conclusion

This experiment demonstrates that deployment is only one stage in the BI lifecycle. Continuous monitoring helps identify operational issues, while planned backups and documented restore procedures reduce disruption when BI assets are lost or damaged. A defined workspace structure, role allocation, and governance process support controlled access and reliable maintenance. Together, these measures improve the availability, reliability, and continuity of BI solutions.